

Statistics are numbers that **summarize raw facts** and **figures** in some meaningful way. They present key ideas that may not be immediately apparent from **the raw data**, and by "**data**" we mean facts or figures from which **we can draw conclusions**.

Categorical or qualitative data

Most of the data we've seen so far is **categorical**. The data is split into categories that describe qualities or characteristics. For this reason, it's also called **qualitative** data. An example of qualitative data is game genre; each genre forms a separate category.

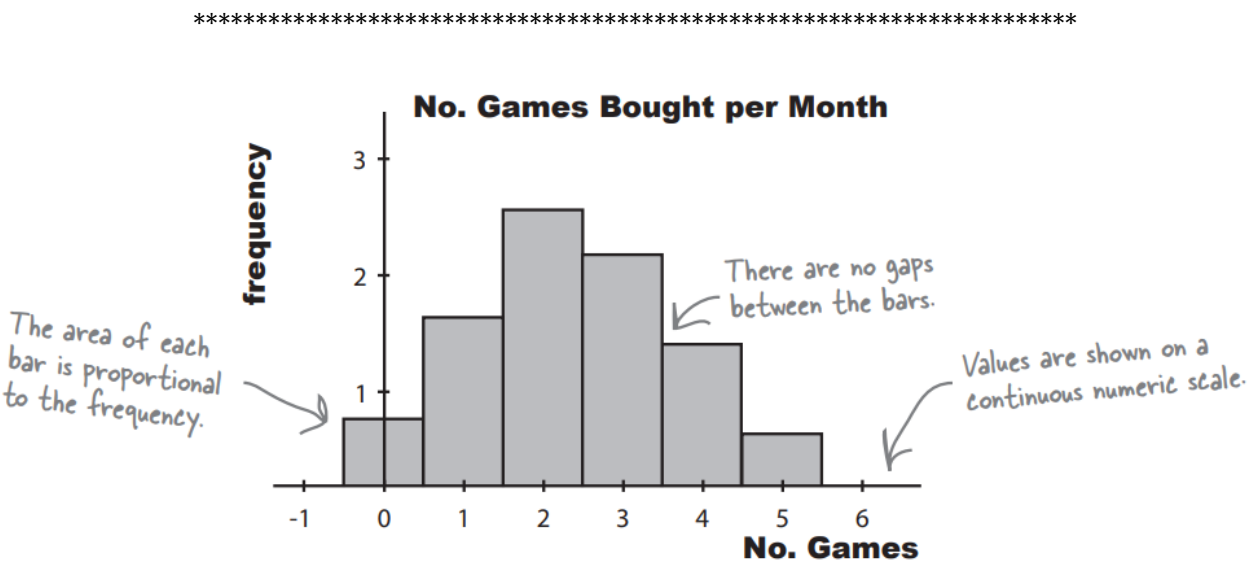
The key thing to remember with qualitative data is that the data values can't be interpreted as numbers.

Numerical or quantitative data

Numerical data, on the other hand, deals with **numbers**. It's data where the values have meaning as numbers, and that involves measurements or counts. Numerical data is also called **quantitative** data because it describes quantities.



Histograms are like bar charts but with two key differences. The first is that the area of each bar is proportional to the frequency, and the second is that there are no gaps between the bars on the chart. Here's an example of a histogram showing the average number of games bought per month by households in Statsville:

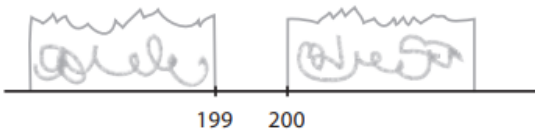


To make a histogram, start by finding bar widths

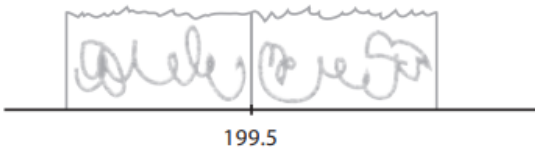
The first step to creating a histogram is to look at each of the intervals and work out how wide each of them needs to be, and what range of values each one needs to cover. While doing this, we need to make sure that there will be no gaps between the bars on the histogram.

Let's start with the first two intervals, 0–199 and 200–399. At face value, the first interval finishes at score 199, and the second starts at score 200. The problem with plotting it like this, however, is that it would leave a gap between score 199 and 200, like this:

Score	Frequency
0–199	5
200–399	29
400–599	56
600–799	17
800–999	3



Histograms shouldn't have gaps between the bars, so to get around this, we extend their ranges slightly. Instead of one interval ending at score 199 and the next starting at score 200, we make the two intervals meet at 199.5, like this:



there are no Dumb Questions

Q: So is a histogram basically for grouped numeric data?

A: Yes it is. The advantage of a histogram is that because its numeric, you can use it to show the width of each interval as well as the frequency.

Q: What about if the intervals are different widths? Can you still use a histogram?

A: Absolutely. It's more common for the interval widths to be equal size, but with a histogram they don't have to be. There are a couple more steps you need to go through to create a histogram with unequal sized intervals, but we'll show you that very soon.

Q: Why shouldn't histograms have gaps between the bars?

A: There are at least two good reasons. The first is to show that there are no gaps in the values, and that every value is covered. The second is so that the width of the interval reflects the range of the values you're covering. As an example, if we drew the interval 0–199 as extending from value 0 to value 199, the width on the chart would only be $199 - 0 = 199$.

Q: So why do we make the bars meet midway between the two?

A: The bars have to meet, and it's usually at the midway point, but it all comes down to how you round your values. When you round values, you normally round them to the nearest whole number. This means that the range of values from -0.5 to 0.5 all round to 0, and so when we show 0 on a histogram, we show it using the range of values from -0.5 to 0.5.

Q: Are there any exceptions to this?

A: Yes, age is one exception. If you have to represent the age range 18–19 on a histogram, you would normally represent this using an interval that goes from 18 to 20. The reason for this is that we typically classify someone as being 19, for example, up until their 20th birthday. In effect, we round ages down.



BULLET POINTS

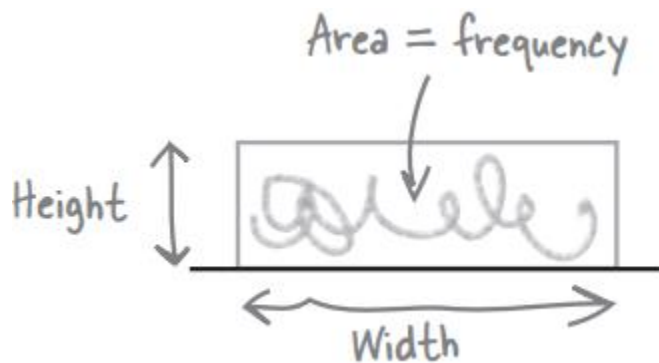
- The **frequency** is a statistical way of saying how many items there are in a category.
- **Pie charts** are good for showing basic proportions.
- **Bar charts** give you more flexibility and precision.
- **Numerical data** deals with numbers and quantities; categorical data deals with words and qualities.
- **Horizontal bar charts** are used for categorical data, particularly where the category names are long.
- **Vertical bar charts** are used for numerical data, or categorical data if the category names are short.
- You can show **multiple sets of data on a bar chart**, and you have a choice of how to do this. You can compare frequencies by showing related bars side-by-side on a split-category bar chart. You can show proportions and total frequencies by stacking the bars on top of each other on a segmented bar chart.
- **Bar chart scales** can show either percentages or frequencies.
- Each chart comes in a number of different varieties.

For histograms,
the frequency is
represented by
bar area

Area of bar = Frequency of group

Frequency = Width of bar × Height of bar

Height of bar = $\frac{\text{Frequency}}{\text{Width of bar}}$



The **height of the bar** is used to measure **how concentrated the frequency is for a particular group**. It's a way of measuring how densely packed the frequency is, a way of saying how thick or thin on the ground the numbers are. The height of the bar **is called the frequency density**.

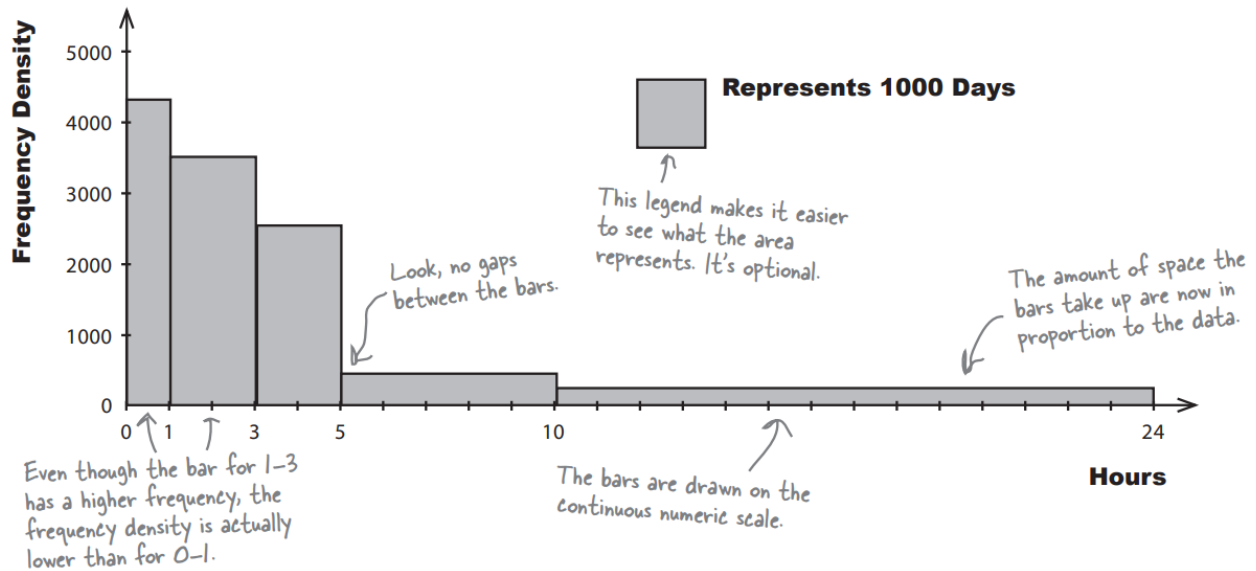


Sharpen your pencil Solution

What should the height of each bar be? Complete the table.

Hours	Frequency	Width	Height (Frequency Density)
0-1	4,300	1	$4,300 \div 1 = 4,300$
1-3	6,900	2	$6,900 \div 2 = 3,450$
3-5	4,900	2	$4,900 \div 2 = 2,450$
5-10	2,000	5	$2,000 \div 5 = 400$
10-24	2100	14	$2,100 \div 14 = 150$

Hours Spent Gaming per Day



Frequency Density Up Close



Frequency density refers to the concentration of values in data. It's related to frequency, but it's not the same thing. Here's an analogy to demonstrate the relationship between the two.

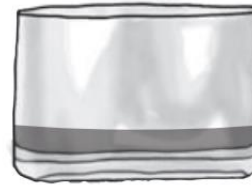
Imagine you have a quantity of juice that you've poured into a glass like this:



Here's all your juice in the glass. It comes up to this level.

What if you then pour the *same quantity* of juice into a *different sized glass*, say one that's wider? What happens to the level of the juice? This time the glass is wider, so the level the juice comes up to is lower.

The level of the juice varies in line with the width of the glass; the wider the glass, the lower the level. The converse is true too; the narrower the glass, the higher the level of juice.



The glass is wider, so the level isn't as high.

bullet points and no dumb questions



BULLET POINTS

- **Frequency density** relates to how concentrated the frequencies are for grouped data. It's calculated using

$$\text{Frequency density} = \frac{\text{Frequency}}{\text{Group width}}$$

- A **histogram** is a chart that specializes in grouped data. It looks like a bar chart, but the height of each bar equates to frequency density rather than frequency.

- When drawing histograms, the width of each bar is proportional to the width of its group. The bars are shown on a continuous numeric scale.
- In a histogram, the frequency of a group is given by the area of its bar.
- A histogram has no gaps between its bars.

Q: Why do we use area to represent frequency when we're graphing histograms?

A: It's a way of making sure the relative sizes of each group stay in proportion to the data, and stay honest. With grouped data, we need a visual way of expressing the width of each group and also its frequency. Changing the width of the bars is an intuitive way of reflecting the group range, but it has the side effect of making some of the bar sizes look disproportionate.

Adjusting the bar height and using the area to represent frequency is a way around this. This way, no group is misrepresented by taking up too much or too little space.

Q: What's frequency density again?

A: Frequency density is a way of indicating how concentrated values are in a particular interval. It gives you a way of comparing different intervals that may be different widths. It makes the frequency proportional to the area of a bar, rather than height.

To find the frequency density, take the frequency of an interval, and divide it by the width.

Q: If I have grouped numeric data, but all the intervals are the same width, can I use a normal bar chart?

A: Using a histogram will better represent your data, as you're still dealing with grouped data. You really want your frequency to be proportional to its area, not height.

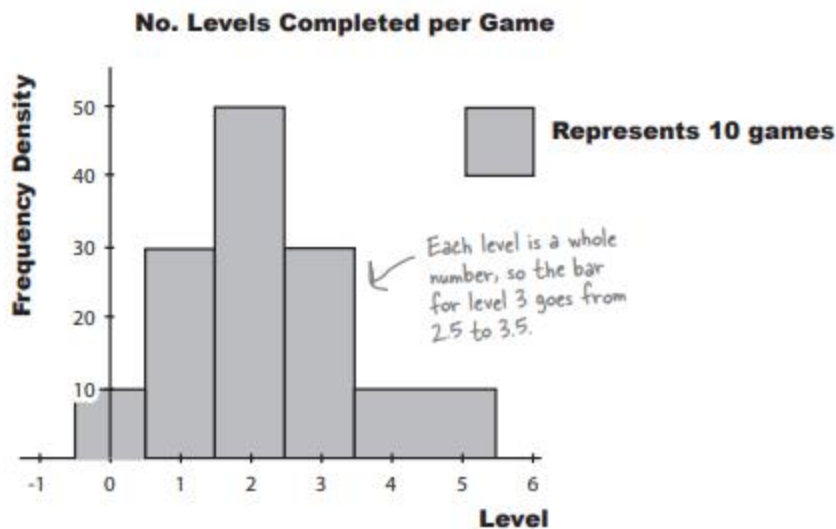
Q: Do histograms *have* to show grouped data? Can you use them for individual numbers as well as groups of numbers?

A: Yes, you can. The key thing to remember is to make sure there are no gaps between the bars and that you make each bar 1 wide. Normally you do this by positioning your number in the center of the bar.

As an example, if you wanted to draw a bar representing the individual number 1, then you'd draw a bar ranging from 0.5 to 1.5, with 1 in the center.



Here's a histogram representing the number of levels completed in each game of Cows Gone Wild. How many games have been played in total? Assume each level is a whole number.

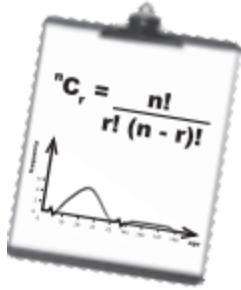


We need to find the total number of games played, which means we need to find the total frequency.

The total frequency is equal to the area of each bar added together. In other words, we multiply the width of each bar by its frequency density to get the frequency, and then add the whole lot up together.

Level	Width	Frequency Density	Frequency
0	1	10	$1 \times 10 = 10$
1	1	30	$1 \times 30 = 30$
2	1	50	$1 \times 50 = 50$
3	1	30	$1 \times 30 = 30$
4-5	2	10	$2 \times 10 = 20$

$$\begin{aligned} \text{Total Frequency} &= 10 + 30 + 50 + 30 + 20 \\ &= 140 \end{aligned}$$



Vital Statistics

Cumulative Frequency

The total frequency up to certain value. It's basically a running total of the frequencies.